Euclid Book I

CGJteamLab

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Contents

Chapter 1

Proposition I.1

Euclid's first proposition in Book I is usually presented as the straightforward construction obtained from the intersection of two circles. A formal reconstruction, however, reveals two logically independent problems: the existence of a common point of the corresponding circles and the proof that the resulting point does not lie on the line determined by the given base.

The proof of non-collinearity relies on the classical idea that a proper part of a segment cannot be congruent to the entire segment. This argument is well known from modern axiomatic reconstructions of geometry, including the work of Michael Beeson and his collaborators.

The purpose of this note is not to present a new geometric proof. Rather, it is to demonstrate that, once Book Zero has been developed, this argument becomes a direct application of the previously established theory. Proposition I.1 therefore serves as the first experimental test of Book Zero as an intermediate language for synthetic proofs.

1.1 The Classical Construction

Let $A \neq B$. Euclid constructs two circles of radius AB: the first centered at A, the second centered at B. One of their points of intersection is denoted by C.

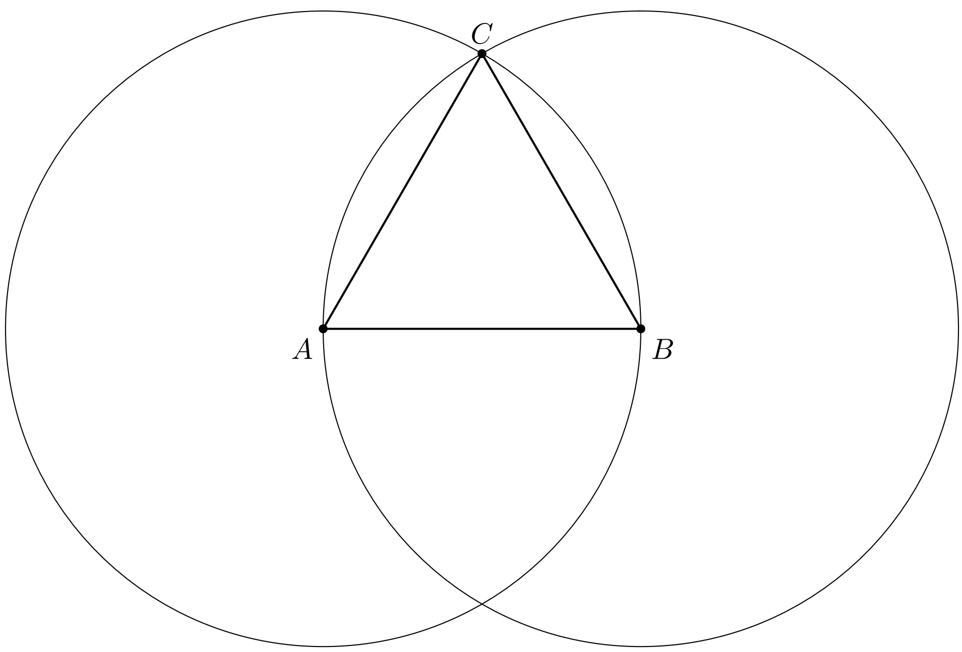


Figure 1.1: Euclid's classical construction. The point C is a common point of two circles of radius AB.

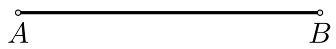
From the construction we obtain

$$AC \cong AB, \qquad BC \cong AB.$$

Formally, however, this is not yet a complete proof. One must also show that the points A, B, C are not collinear.

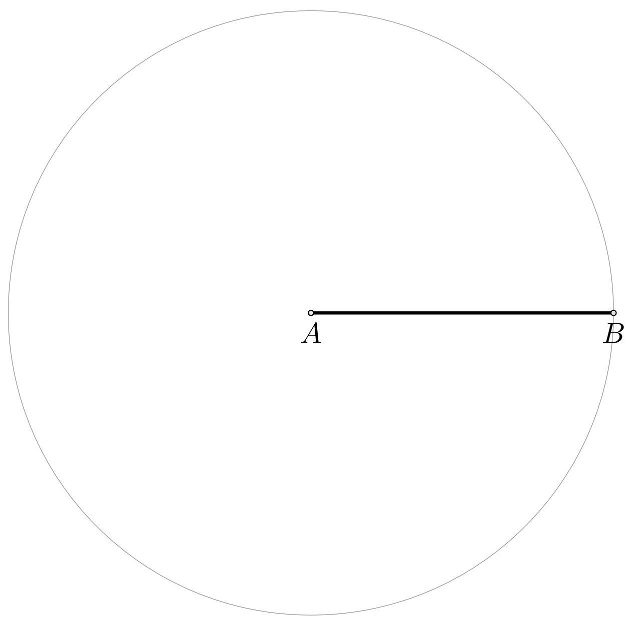
Step 1. Given Segment

The construction starts with a given segment AB.



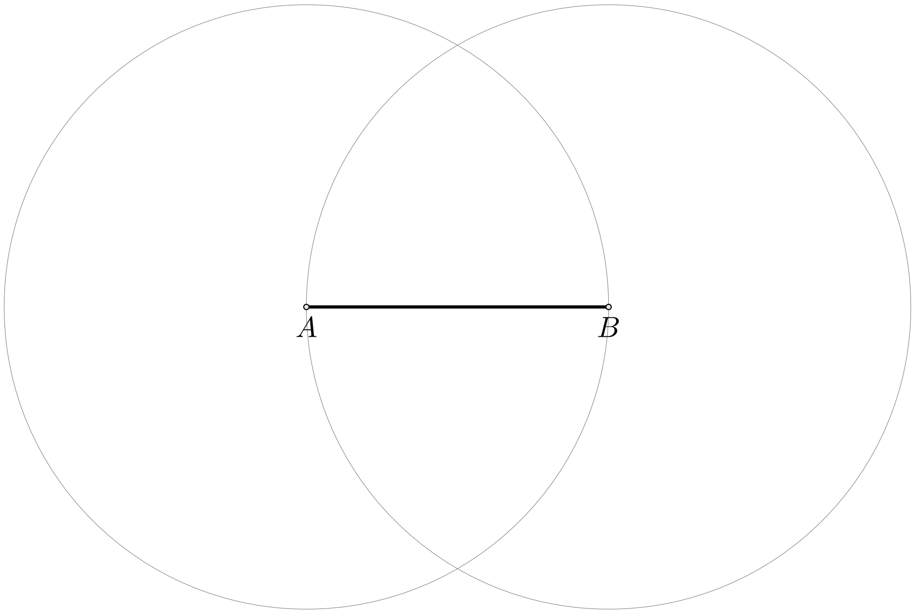
Step 2. Draw the Circle Centered at A

Draw the circle centered at A through B.



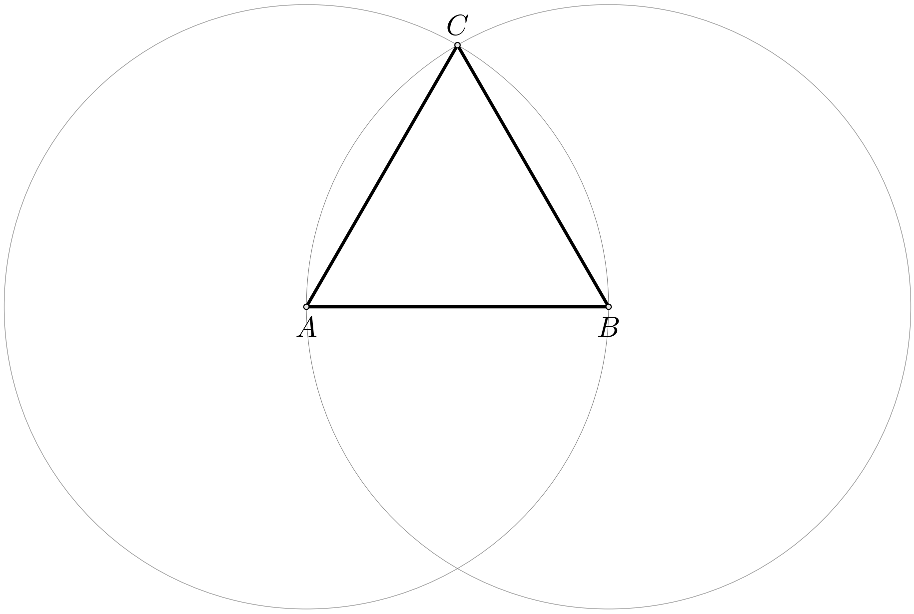
Step 3. Draw the Circle Centered at B

Draw the circle centered at B through A.



Step 4. Construct the Equilateral Triangle

Let C be one of the intersection points of the two circles. Join AC and BC to obtain the equilateral triangle.



1.2 Separating the Two Problems

The reconstruction reveals two logically independent stages.

1. **Existence stage.** One must obtain a point C such that

$$AC \cong AB, \qquad BC \cong AB.$$

   In the current version of the project, this stage is provided by the temporary principle `hilbert_equidistant_point_exists`. Ultimately, it is intended to be derived from an appropriate theory of continuity in Hilbert geometry.

2. **Synthetic stage.** From the congruences above one must derive

$$\neg\,\mathrm{Collinear}(A, B, C).$$

   This stage no longer requires any theory of circles.

1.3 Theorem

Theorem. Let $A \neq B$. Assume that there exists a point C such that

$$AC \cong AB, \qquad BC \cong AB.$$

Then the points A, B, C are not collinear.

1.4 Proof

First observe that $C \neq A$ and $C \neq B$. If $C = A$, then $AC \cong AB$ would imply that a null segment is congruent to the non-null segment AB. Similarly, $C \neq B$.

Now suppose, for contradiction, that the points A, B, C are collinear. Since they are pairwise distinct, the trichotomy of the order of points on a line gives exactly three possibilities:

$$A - B - C, \qquad B - A - C, \qquad A - C - B.$$

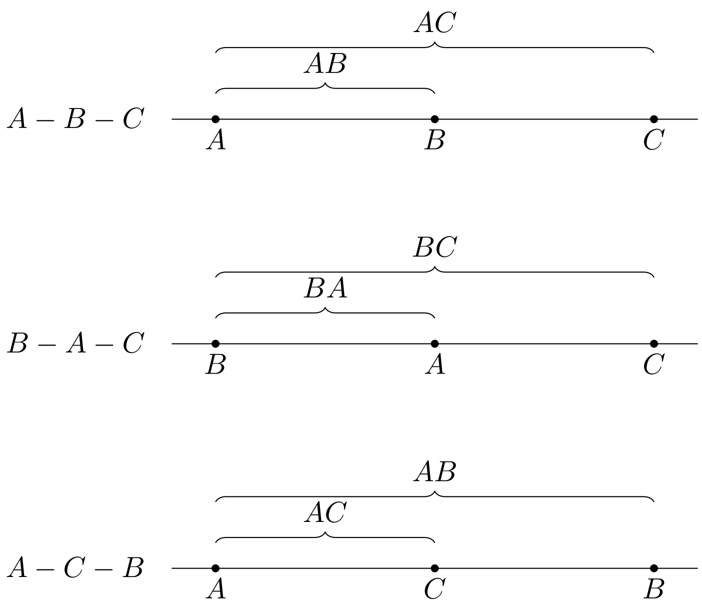


Figure 1.2: The three possible arrangements of pairwise distinct collinear points.

Case 1: $A - B - C$

The segment AB is a proper part of the segment AC. Book Zero contains the general theorem:

> A proper part of a segment cannot be congruent to the whole segment.

On the other hand, by assumption,

$$AC \cong AB.$$

Using symmetry of congruence, we obtain

$$AB \cong AC,$$

which is a contradiction.

Case 2: $B - A - C$

The segment BA is a proper part of the segment BC. Since

$$BC \cong AB,$$

and reversing the endpoints of a segment does not change its length, we have

$$BA \cong AB.$$

By transitivity of congruence, it follows that

$$BA \cong BC,$$

again contradicting the fact that a proper part cannot be congruent to the whole.

Case 3: $A - C - B$

The segment AC is a proper part of the segment AB. But the assumption directly gives

$$AC \cong AB.$$

This yields the third contradiction.

All possible collinear arrangements lead to a contradiction. Therefore,

$$\neg\,\mathrm{Collinear}(A, B, C).$$

□

1.5 Architecture of the Proof

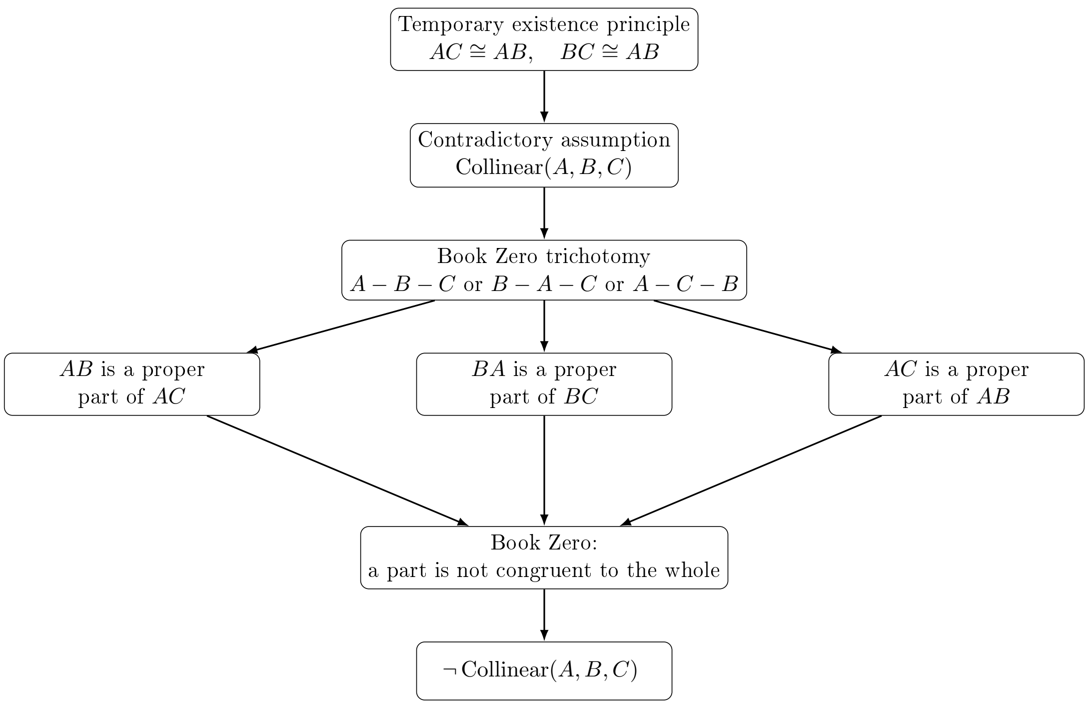


Figure 1.3: The structure of the proof of Proposition I.1 above the Book Zero layer.

1.6 Significance of the Reconstruction

The geometric result itself is not new. Constructing an equilateral triangle on a given segment is one of the classical theorems of elementary geometry.

Likewise, the idea underlying the proof of non-collinearity is not new. Modern axiomatic reconstructions, in particular the work of Michael Beeson and his collaborators, employ an analogous argument based on the theorem that a proper part of a segment cannot be congruent to the entire segment.

The novelty of the present reconstruction lies elsewhere.

The objective of the CGJteamLab project was not to discover a new proof of Proposition I.1, but to investigate whether the theory developed in Book Zero is sufficiently rich to serve as an intermediate language for synthetic proofs.

The experiment shows that, once the existence problem for an equidistant point has been isolated, the entire argument can be expressed almost exclusively in terms of the theorems previously established in Book Zero.

From the perspective of the architecture of the theory, this is more significant than Proposition I.1 itself. It demonstrates that Book Zero is not merely a collection of auxiliary lemmas, but rather forms a new layer of abstraction between Hilbert's axioms and classical synthetic reasoning.

Schematically, the situation may be represented as

$$\text{Hilbert's axioms} \longrightarrow \text{Book Zero} \longrightarrow \text{Proposition I.1.}$$

In this sense, Proposition I.1 constitutes the first experimental test confirming that Book Zero can serve as a language of synthetic proofs for the further reconstruction of Euclidean geometry.

Chapter 2

Proposition I.2

This note compares Euclid's original construction for Proposition I.2 with its reconstruction over the Hilbert foundation. In the CGJteamLab development the proposition follows immediately from Book Zero #49 (Layoff), illustrating the difference between the architectural roles of segment construction in Euclid's geometry and the Hilbert axiomatic framework.

2.1 Statement

Given a point A and a segment BC, construct a point D such that

$$AD \cong BC.$$

2.2 Euclid's Classical Construction

Euclid's construction proceeds as follows.

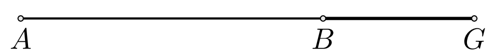
1. Construct the equilateral triangle DAB on the segment AB.
2. Produce the side DB beyond B.
3. Produce the side DA beyond A.
4. Draw the circle centered at B with radius BC.
5. Draw the circle centered at D with radius DG.
6. The constructed point L satisfies
$$AL \cong BC.$$

Step 1. Given data

The construction starts with a given point A and a given segment BG.

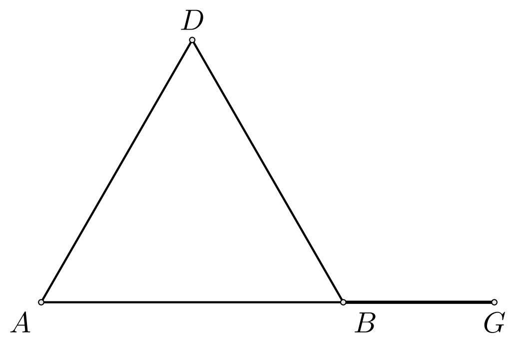
Step 2. Join AB

Join the given point A to the endpoint B of the given segment.



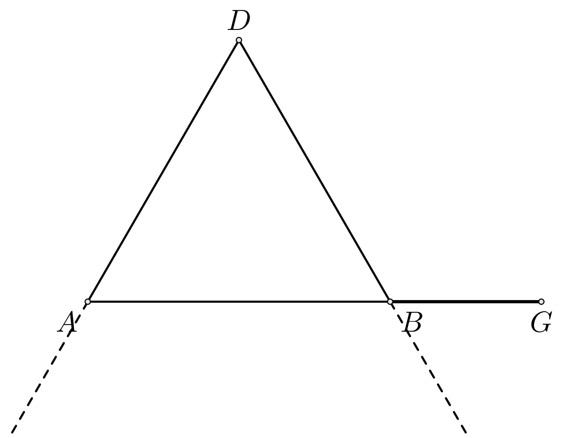
Step 3. Construct the equilateral triangle DAB

Construct the equilateral triangle DAB on the segment AB.



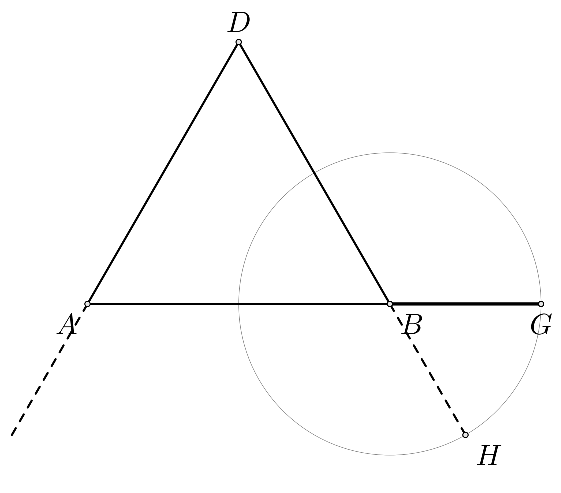
Step 4. Produce DA and DB

Produce DA beyond A and DB beyond B.



Step 5. Draw the circle centered at B

Draw the circle centered at B through G. Let H be the intersection of this circle with the line DB on the side beyond B.

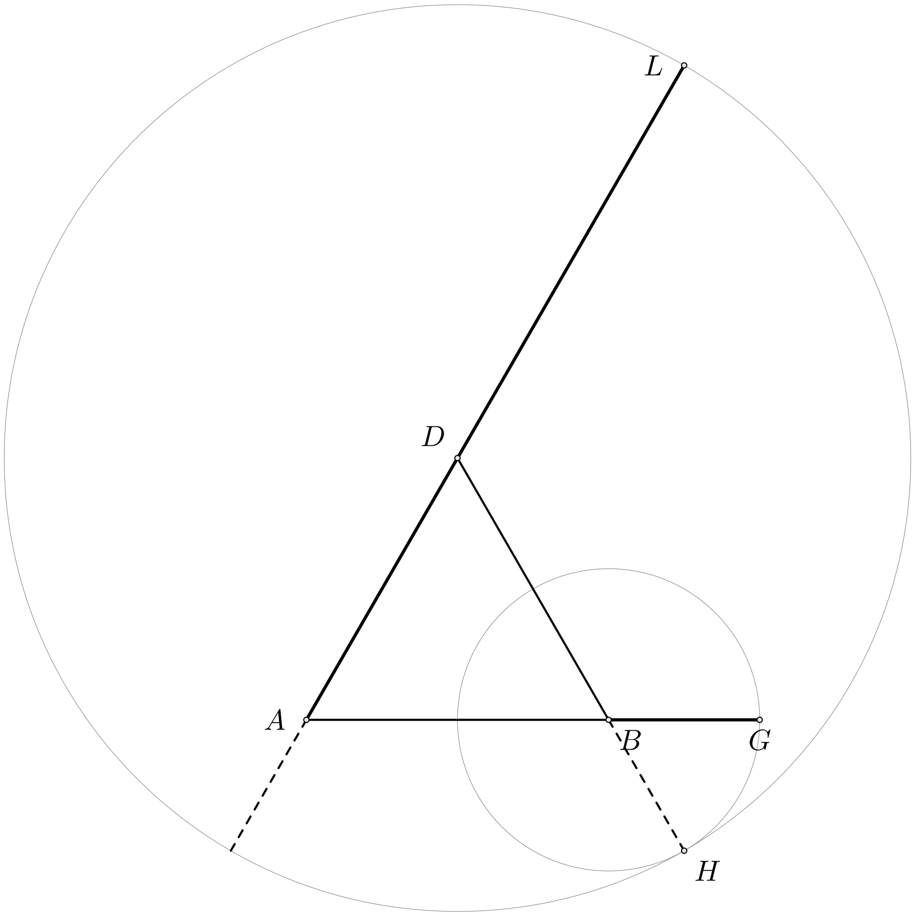


Since H lies on the circle centered at B through G,

$$BH \cong BG.$$

Step 6. Draw the circle centered at D

Draw the circle centered at D through H. Let L be the intersection of this circle with the line DA on the side beyond A.



Since L lies on the circle centered at D through H,

$$DL \cong DH.$$

The segment AL is therefore the required copy of the given segment.

2.3 Hilbert Reconstruction

The reconstruction over the Hilbert foundation is considerably shorter.

1. Choose any point $E \neq A$.
2. The ray AE is thereby determined.
3. Apply Book Zero #49 (Layoff).
4. Obtain a point D on the ray AE satisfying

$$AD \cong BC.$$

No auxiliary geometric construction is required.
Book Zero #49 provides a point D on a chosen ray from A such that

$$AD \cong BC.$$

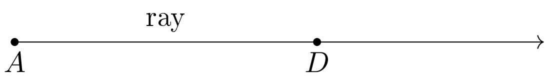


Figure 2.1: Lay off the segment on a chosen ray.

2.4 Formalization

The formal proof consists of a single application of Book Zero #49.

```
theorem euclid_proposition_2 ...
  := bookZero_49_layoff ...
```

2.5 Discussion

Thus Proposition I.2 illustrates a fundamental architectural difference between Euclid's geometry and the Hilbert approach.

In Euclid's *Elements*, the transport of a segment is obtained through a sequence of explicit geometric constructions beginning with Proposition I.1.

In contrast, Hilbert incorporates this operation into the congruence axioms. Book Zero encapsulates it in the Layoff Theorem, making Proposition I.2 an immediate consequence.

2.6 Book Zero #49: Layoff

The proof of Proposition I.2 relies on one of the elementary construction theorems established in Book Zero.

Book Zero #49 (Layoff). Let $A \neq B$ and $C \neq D$. Then there exists a point X on the ray AB such that

$$AX \cong CD.$$

In other words, every nondegenerate segment can be copied onto an arbitrarily chosen ray while preserving its length.

```
Input
-----

Ray AB

A----------------------------->

Segment CD

C-------------D


|
|  Book Zero #49
v


Output
```

```
------

A-----------------X------------>

AX congruent CD
```

This theorem provides the fundamental operation of transporting segments. Once it is available, Euclid's Proposition I.2 becomes an immediate consequence.

Bibliography

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[5] Michael Beeson and Larry Wos, "Axiomatizing Geometry," *Annals of Mathematics and Artificial Intelligence*, 2019.